

Presentation

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The development of stratification and segregation in a new scientific field – A study of collaboration among scientists in neuroblastoma research between 1975 and 2016

Mark Wittek¹, Christoph Bartenhagen², Frank Berthold²

¹University of Cologne, Institute of Sociology and Social Psychology ²University of Cologne, Department of Pediatric Oncology and Hematology, Germany

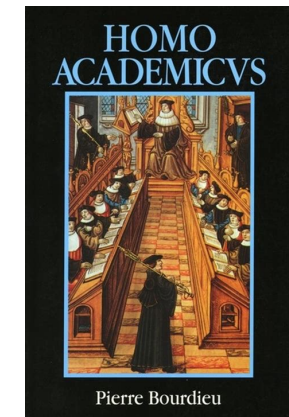
Background

- ▶ Sociology of Science highlights the importance of social processes for scientific collaboration and knowledge production (Kuhn 1962; Merton 1973; Bourdieu 1988; Latour 1987, Crane 1972)
- ▶ Scientific work is marked by inequality in social recognition (Newman 2001; Eom & Fortunato 2011), segregation by organizational status (Ma et al. 2020), and unequal access to resources such as funding (Bol et al. 2018)
- ▶ We investigate the scientific field of neuroblastoma research to study the development of stratification and segregation in scientific collaboration networks



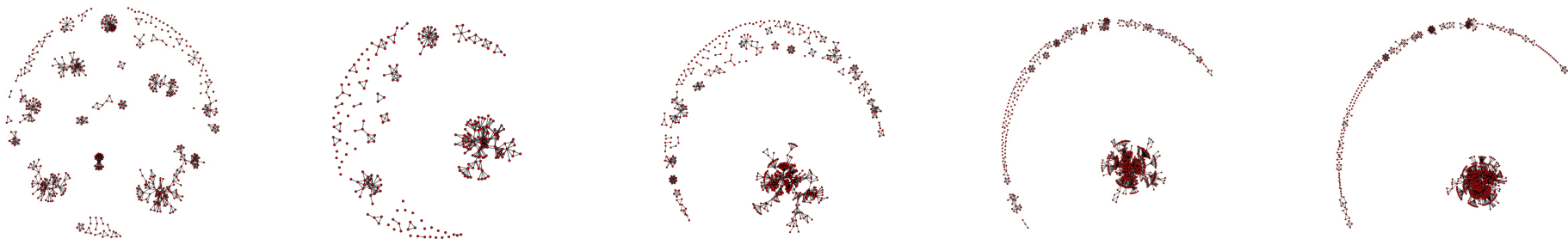
World Rank	Institution*	Country /Region	World Rank	Institution*	Country /Region
1	Harvard University		11	Yale University	
2	Stanford University		12	University of California, Los Angeles	
3	Massachusetts Institute of Technology (MIT)		13	Cornell University	
4	University of California, Berkeley		14	University of California, San Diego	
5	University of Cambridge		15	University of Washington	
6	Princeton University		16	Johns Hopkins University	
7	California Institute of Technology		17	University of Pennsylvania	
8	Columbia University		18	University College London	
9	University of Chicago		18	University of California, San Francisco	
10	University of Oxford		20	Swiss Federal Institute of Technology Zurich	

SOURCE: THE ACADEMIC RANKING OF WORLD UNIVERSITIES (ARWU)



Research Gap

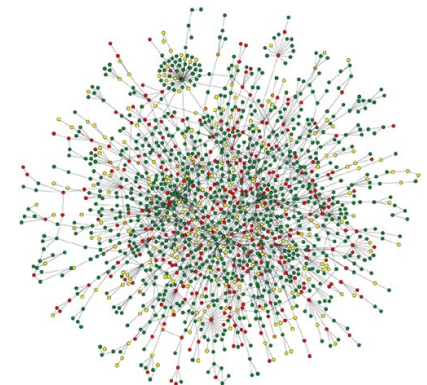
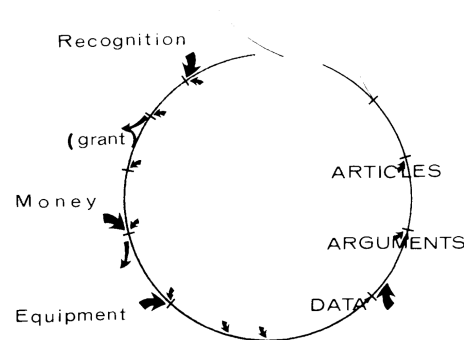
- ▶ Previous studies report inequality in collaboration partners (Newman 2001) and illustrate that the number and prestige of collaboration partners are crucial for individuals' careers (Li et al. 2019). Yet, empirical accounts seldom investigate how inequality in science evolves in the first place and changes over time
- ▶ Application of network models to scientific networks still rare (cf., Kronegger et al. 2021), convergence issues, lacking comparability of estimates, mechanical link between size and network measures (Anderson et al. 1999)
- ▶ We overcome these issues by using recent advances in network analysis (Duxbury 2021) and studied neuroblastoma research from its genesis onward



Theory – Inequality

- ▶ Researchers channel attention on a small set of others: you cannot read, meet or collaborate with everyone, selection is unavoidable (see Fiske 2011, Dunbar 2008; Podolny 2010)
- ▶ Even if a author is only slightly better known than others this will greatly enhance her chance to attract more and more attention (Matthew effect, Merton 1968)
- ▶ New fields lack resources and legitimization, cycles of accumulation hard to initiate (Latour and Woolgar 1986)

H1: Inequality in the distribution of coauthorship ties increases as a scientific field matures and grows.



Theory – Accumulation dynamics and homogeneity

- ▶ Previous accounts described the temporal unfolding of research communities and observed that actors can draw upon multiple field-specific resources to mobilize additional collaborators as fields mature (Chubin, 1976; Frickel and Gross, 2005; Griffith and Mullins, 1972).
- ▶ Status theories suggest that status signals diversify as uncertainty linked to field size increases (Mayhew 1976, Podolny 2010)

H2: Accumulation dynamics diversify as a scientific field matures and grows.

- ▶ Elites develop due to the accumulation of resources and recognition, thereby the status of collaborators becomes more important and increasingly overpowers the tendency for local, status similar collaboration (Blau 1994; Burris 2004; Ma et al. 2020)

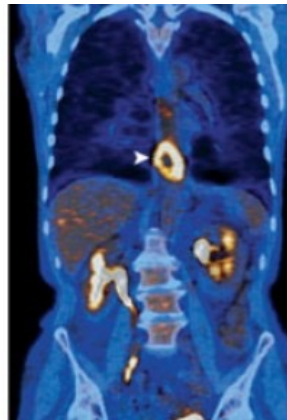
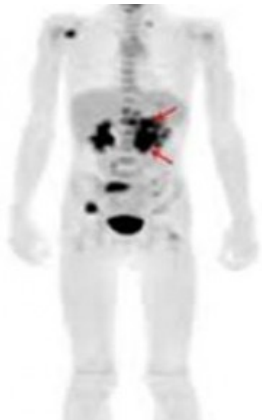
H3: Scientists who are dissimilar in their years of experience, productivity, and seniority collaborate at the outset of a field. This tendency weakens in the later stages of a field's development, and collaboration increasingly exhibits status homogeneity.

Empirical setting

- ▶ Neuroblastoma—most frequent solid tumor in childhood —global research collaborations recorded at Advances in Neuroblastoma Research conferences (ANR); (Berthold et al. 2019)
- ▶ Record of all abstracts handed in to 18 conferences in a time span of 41 years (1975-2016)

Advantages of the data:

1. Emergence of interdisciplinary research field organized around a clear-cut research topic
2. Conference co-authorships as indicator of rather substantial collaboration
3. Conferences as events are important for the evolution of a scientific field



Data and Measures

Data

8459 authors affiliated to 553 institutions of 53 countries presented 3,993 abstracts. The number of coauthors increased over the years from 2 to 7 (Berthold et al. 2019).

Measures

Networks: collaboration networks derived from abstracts handed in to the ANR conference series. We included only actors who appeared on at least two abstracts at two different conferences (ca. 40% of the initial sample)

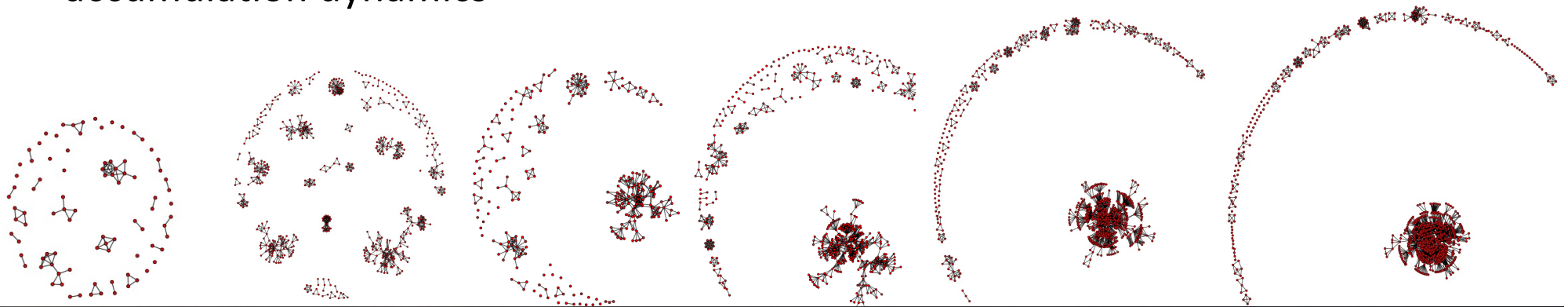
Experience: years since entering the field

Productivity: publication record in neuroblastoma research (Web of science database)

Seniority: share of last author publications

Analytical Strategy

- ▶ Skewness of degree distribution in combination with network simulations from basic and full ERGMs to study changes in inequality and to rule out methodological artifacts connected with network size (Anderson et al. 1999)
- ▶ Exponential random graph models and average marginal effects (Lusher et al. 2013; ERGMs; Duxburry 2021) to capture how collaboration is shaped by accumulation dynamics



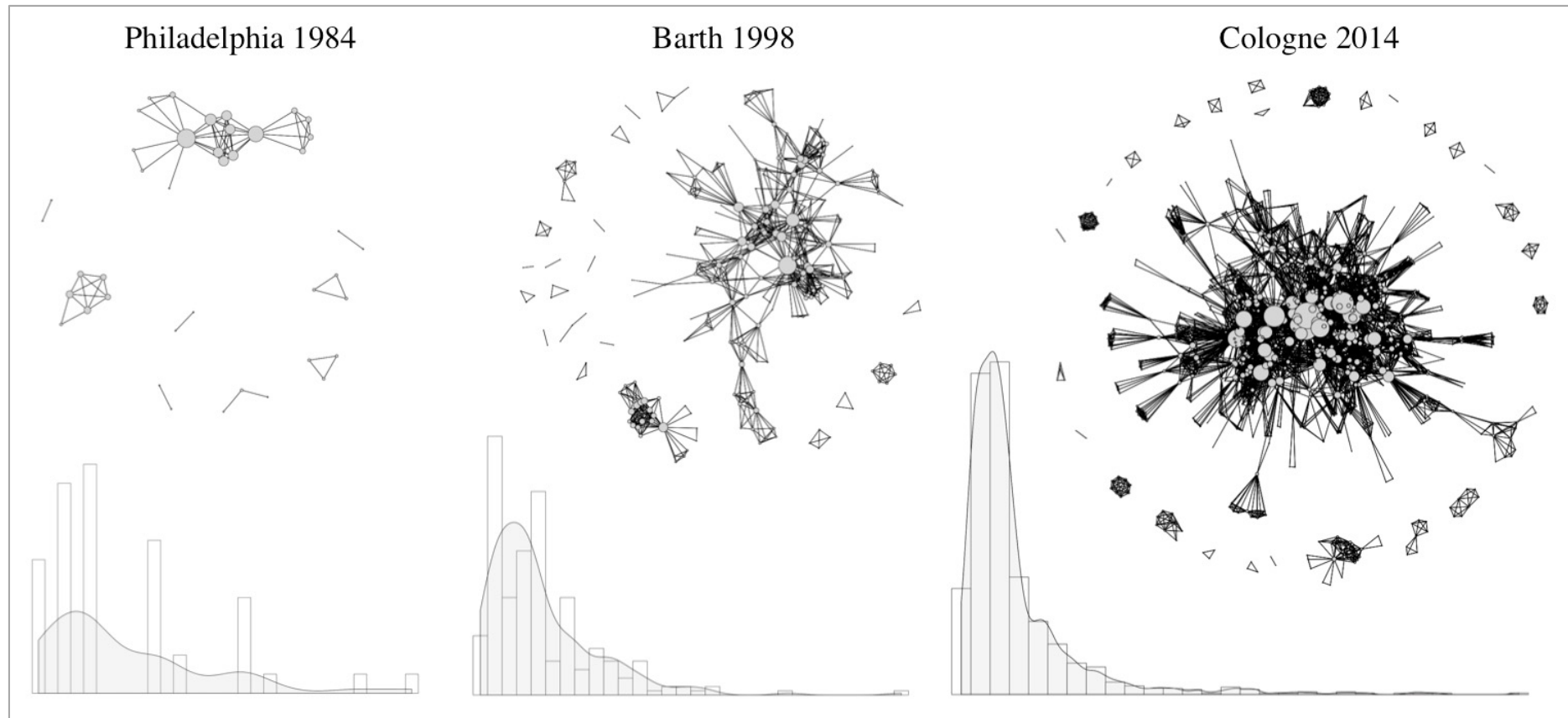
ERGMs Specification

- ▶ We considered the role of experience, productivity, and seniority and controlled for institutional affiliation and country
- ▶ ERGMs control for network density and opportunity structure by default
- ▶ More complex specifications including geometrically weighted terms for e.g., transitivity showed convergence issues

Descriptive Table Time Trends

Year	City of congress	Country of congress	Size	Average degree centrality	Skewness of degree distribution	Standard deviation of degree distribution
1975	Philadelphia	USA	13	0.50	1.91	0.38
1979	Philadelphia	USA	36	1.92	0.04	1.00
1984	Philadelphia	USA	97	6.00	1.76	4.20
1987	Philadelphia	USA	114	5.82	1.63	6.70
1990	Philadelphia	USA	174	5.75	1.58	4.52
1993	Philadelphia	USA	262	8.84	2.17	8.03
1994	Philadelphia	USA	205	10.13	1.25	4.12
1996	Philadelphia	USA	470	12.89	3.30	11.07
1998	Bath	UK	507	11.75	2.56	6.88
2000	Philadelphia	USA	488	12.47	2.81	8.66
2002	Paris	France	641	11.96	2.83	7.30
2004	Genoa	Italy	920	12.66	2.73	7.20
2006	Los Angeles	USA	908	14.12	3.47	8.45
2008	Chiba	Japan	970	19.51	3.12	13.84
2010	Stockholm	Sweden	1184	21.90	3.23	16.85
2012	Toronto	Canada	1117	17.00	3.32	9.93
2014	Cologne	Germany	1197	23.27	3.32	16.64
2016	Cairns	Australia	955	26.69	2.98	19.29

Inequality in Collaboration Networks over Time



ERGMs baseline models

<i>Years</i>	1979		1984		1987		1990		1993		1994	
	AME	Scaled AME	AME	Scaled AME	AME	AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	-0.004 (0.021)	-0.094	0.094*** (0.008)	1.793	0.061*** (0.006)	0.996	0.075*** (0.003)	2.273	0.049*** (0.002)	1.594	0.080*** (0.003)	2.441
Same institution	0.110*** (0.017)	2.754	0.091*** (0.008)	1.739	0.095*** (0.009)	1.548	0.053*** (0.003)	1.597	0.055*** (0.003)	1.768	0.030*** (0.003)	0.931
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	

<i>Years</i>	1996		1998		2000		2002		2004		2006	
		Scaled		Scaled		Scaled		Scaled		Scaled		Scaled
	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME
Same country	0.039*** (0.001)	1.895	0.034*** (0.001)	2.121	0.030*** (0.001)	1.677	0.031*** (0.001)	2.276	0.022*** (0.0004)	2.347	0.020*** (0.0004)	1.911
Same institution	0.034*** (0.001)	1.656	0.027*** (0.001)	1.721	0.035*** (0.001)	1.971	0.025*** (0.001)	1.877	0.020*** (0.0003)	2.107	0.024*** (0.0003)	2.291
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	

<i>Years</i>	2008		2010		2012		2014		2016	
		Scaled		Scaled		Scaled		Scaled		Scaled
	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME
Same country	0.021*** (0.0004)	1.704	0.021*** (0.0003)	1.775	0.018*** (0.0003)	2.012	0.023*** (0.0003)	1.881	0.029*** (0.0004)	1.630
Same institution	0.026*** (0.0004)	2.052	0.025*** (0.0003)	2.089	0.021*** (0.0003)	2.291	0.026*** (0.0003)	2.089	0.036*** (0.0005)	1.980
Baseline probability	0.013		0.011		0.009		0.012		0.018	

ERGMs full models I

<i>Years</i>	1979	1984		1987		1990		1993		1994		
		Scaled		Scaled		Scaled		Scaled		Scaled		Scaled
	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME	AME
Same country	-0.004 (0.021)	-0.111	0.097*** (0.008)	1.836	0.065*** (0.006)	1.062	0.074*** (0.003)	2.258	0.045*** (0.002)	1.463	0.08*** (0.003)	2.353
Same institution	0.105*** (0.017)	2.643	0.091*** (0.008)	1.720	0.090*** (0.009)	1.470	0.054*** (0.003)	1.648	0.060*** (0.003)	1.938	0.03*** (0.003)	1.038
Popularity years of experience	0.001 (0.006)	0.034	-0.009** (0.003)	-0.177	0.006* (0.003)	0.091	0.002 (0.001)	0.065	0.005*** (0.001)	0.159	0.002 (0.001)	0.057
Difference in authors' years of experience	-0.007 (0.009)	-0.174	-9e-03 (0.004)	-0.018	-0.007* (0.003)	-0.118	-0.005** (0.002)	-0.140	-0.004*** (0.001)	-0.138	-0.005** (0.002)	-0.147
Popularity cumulated publications	0.008 (0.011)	0.206	0.007 (0.006)	0.128	0.014*** (0.003)	0.227	0.007*** (0.001)	0.198	0.009*** (0.001)	0.278	0.005** (0.001)	0.168
Difference in cumulated publications	-0.024 (0.017)	-0.595	0.005 (0.007)	0.085	0.006 (0.003)	0.093	0.002 (0.002)	0.064	-0.002* (0.001)	-0.079	-0.001 (0.002)	-0.038
Popularity share last author positions	0.001 (0.007)	0.028	-0.008 (0.006)	-0.160	-0.013** (0.005)	-0.210	-0.015*** (0.003)	-0.466	-0.005*** (0.001)	-0.170	-0.01*** (0.002)	-0.243
Difference in share last author positions	0.003 (0.009)	0.067	0.008 (0.006)	0.190	0.004 (0.006)	0.072	0.013*** (0.003)	0.389	0.001 (0.002)	0.039	0.007** (0.002)	0.200
Baseline probability	0.040		0.053		0.062		0.033		0.031		0.032	

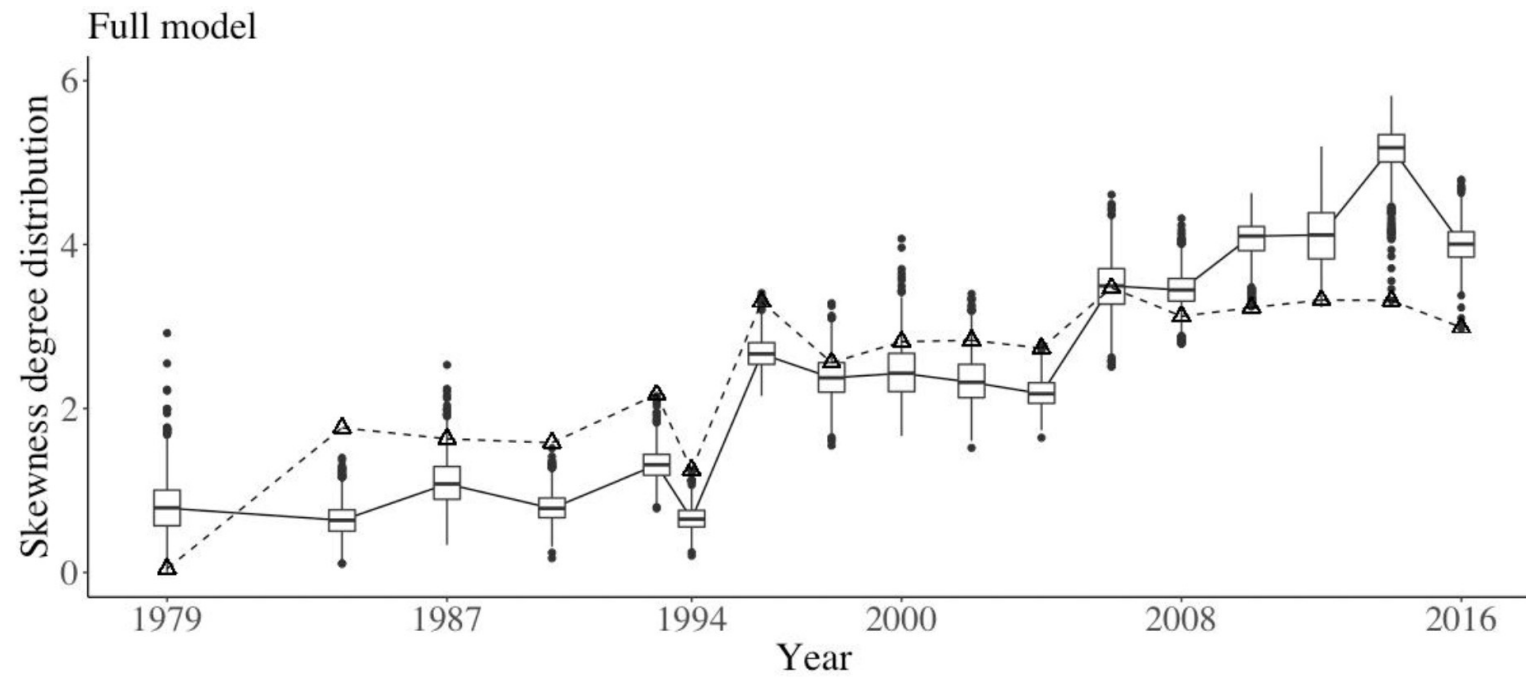
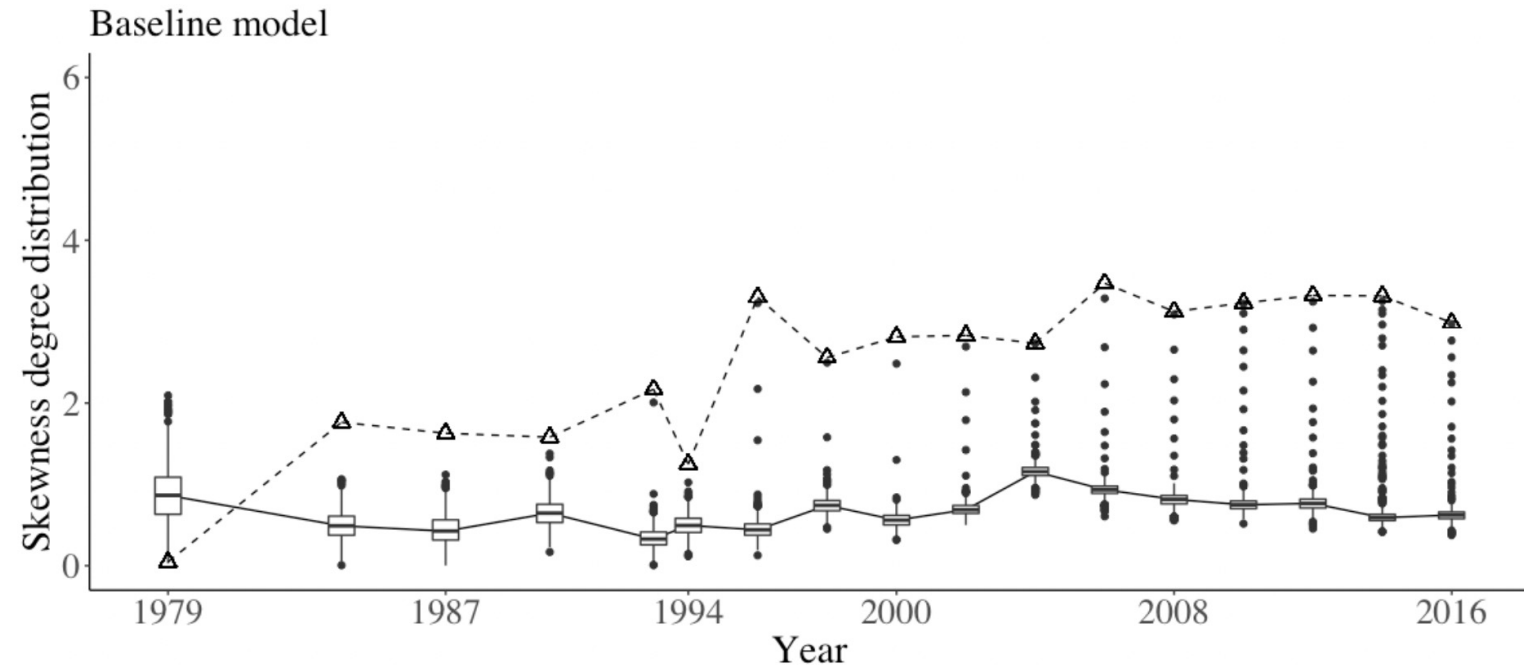
ERGMs full models II

<i>Years</i>	1996		1998		2000		2002		2004		2006	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.037*** (0.001)	1.802	0.032*** (0.001)	2.035	0.029*** (0.001)	1.616	0.029*** (0.001)	2.184	0.021*** (0.0004)	2.280	0.019*** (0.0004)	1.853
Same institution	0.036*** (0.001)	1.744	0.028*** (0.001)	1.770	0.036*** (0.001)	1.990	0.025*** (0.001)	1.897	0.020*** (0.0003)	2.090	0.023*** (0.0003)	2.262
Popularity years of experience	0.0001 (0.0003)	0.006	0.0007* (0.0003)	0.044	0.002*** (0.0003)	0.126	0.002*** (0.0002)	0.139	0.001*** (0.0001)	0.086	0.001*** (0.0001)	0.120
Difference in authors' years of experience	-0.004*** (0.001)	-0.184	-0.005*** (0.0004)	-0.340	-0.004*** (0.0004)	-0.243	-0.003*** (0.0003)	-0.213	-0.002*** (0.0002)	-0.164	-0.002*** (0.0002)	-0.178
Popularity cumulated publications	0.010*** (0.0003)	0.462	0.006*** (0.0003)	0.353	0.006*** (0.0004)	0.334	0.003*** (0.0003)	0.216	0.002*** (0.0002)	0.195	0.003*** (0.0002)	0.305
Difference in cumulated publications	-0.002*** (0.0004)	-0.082	>0.0001 (0.0004)	0.016	-0.002*** (0.0004)	-0.081	-0.0004 (0.0003)	-0.020	0.0002 (0.0002)	0.017	-0.001*** (0.0002)	-0.069
Popularity share last author positions	-0.003*** (0.001)	-0.145	-0.002*** (0.001)	-0.151	-0.003*** (0.001)	-0.190	-0.0003 (0.0003)	-0.024	-0.001*** (0.0002)	-0.085	-0.001*** (0.0002)	-0.121
Difference in share last author positions	>0.001 (0.001)	0.026	0.002*** (0.001)	0.128	0.003*** (0.001)	0.160	0.001* (0.0004)	0.064	0.001** (0.0002)	0.068	0.001* (0.0003)	0.066
Baseline probability	0.021		0.016		0.018		0.013		0.009		0.010	

ERGMs full models III

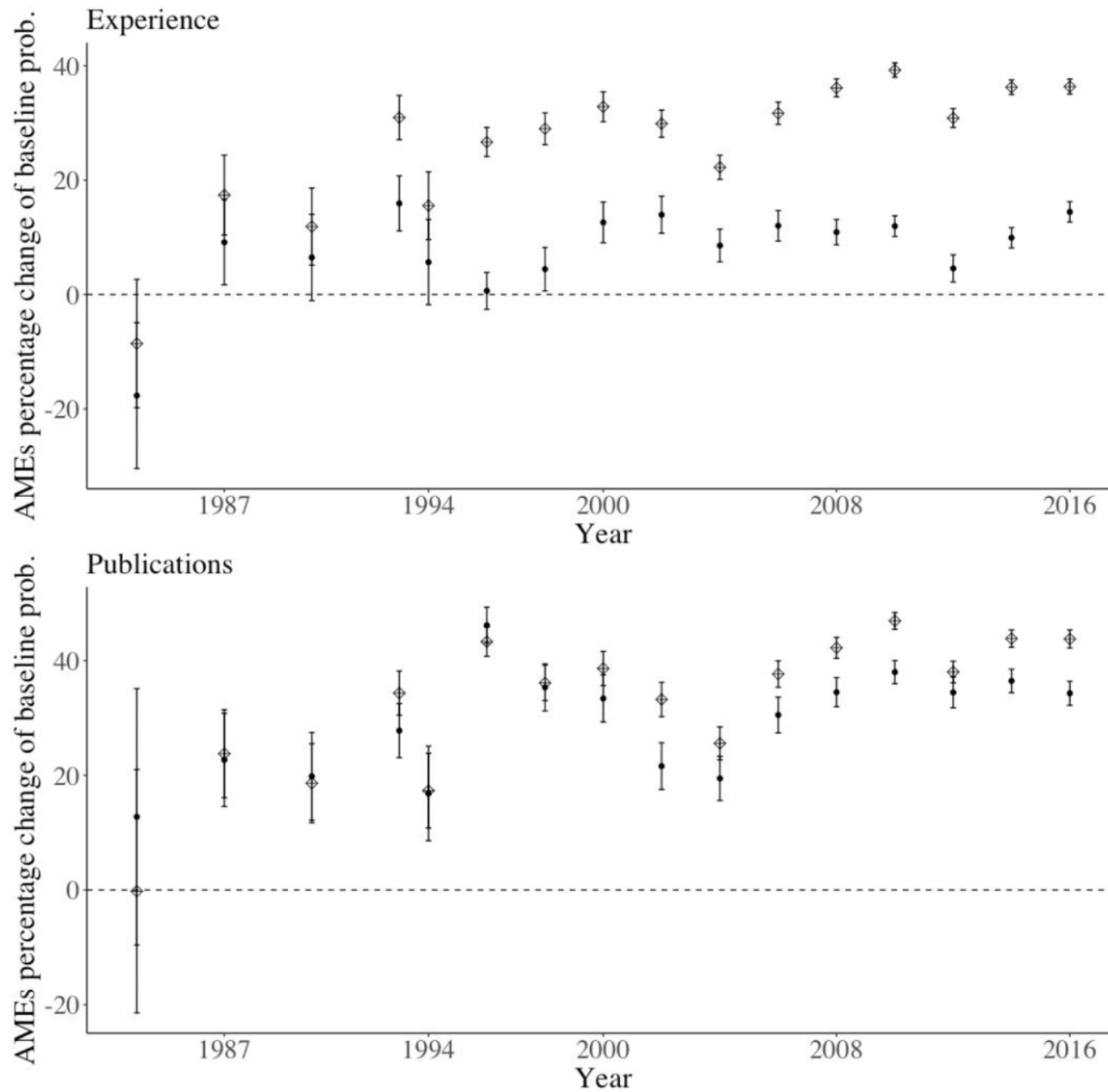
<i>Years</i>	2008		2010		2012		2014		2016	
	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME	AME	Scaled AME
Same country	0.020*** (0.0004)	1.620	0.019*** (0.0003)	1.633	0.017***	1.894	0.022*** (0.0003)	1.794	0.028*** (0.0004)	1.571
Same institution	0.026*** (0.0004)	2.072	0.025*** (0.0003)	2.163	0.021***	2.295	0.026*** (0.0003)	2.115	0.037*** (0.001)	2.020
Popularity years of experience	0.001*** (0.0001)	0.109	0.001*** (0.0001)	0.120	0.0004***	0.046	0.001*** (0.0001)	0.099	0.003*** (0.0002)	0.145
Difference in authors' years of experience	-0.002*** (0.0002)	-0.178	-0.002*** (0.0002)	-0.187	-0.001***	-0.114	-0.002*** (0.0002)	-0.129	-0.003*** (0.0002)	-0.142
Popularity cumulated publications	0.004*** (0.0002)	0.345	0.005*** (0.0001)	0.380	0.003***	0.345	0.005*** (0.0001)	0.365	0.006*** (0.0002)	0.343
Difference in cumulated publications	-0.001*** (0.0002)	-0.072	-0.002*** (0.0001)	-0.131	-0.001***	-0.084	-0.001*** (0.0001)	-0.104	-0.002*** (0.0002)	-0.110
Popularity share last author positions	-0.001*** (0.0002)	-0.080	-0.0004* (0.0002)	-0.045	-0.0004*	-0.060	-0.0004* (0.0002)	-0.031	-0.002*** (0.0003)	-0.102
Difference in share last author positions	-0.0004 (0.0003)	-0.015	-0.0005* (0.0002)	-0.042	0.0003	0.035	-0.0005* (0.0002)	-0.043	-0.001 (0.0004)	-0.038
Baseline probability	0.013		0.011		0.009		0.012		0.018	

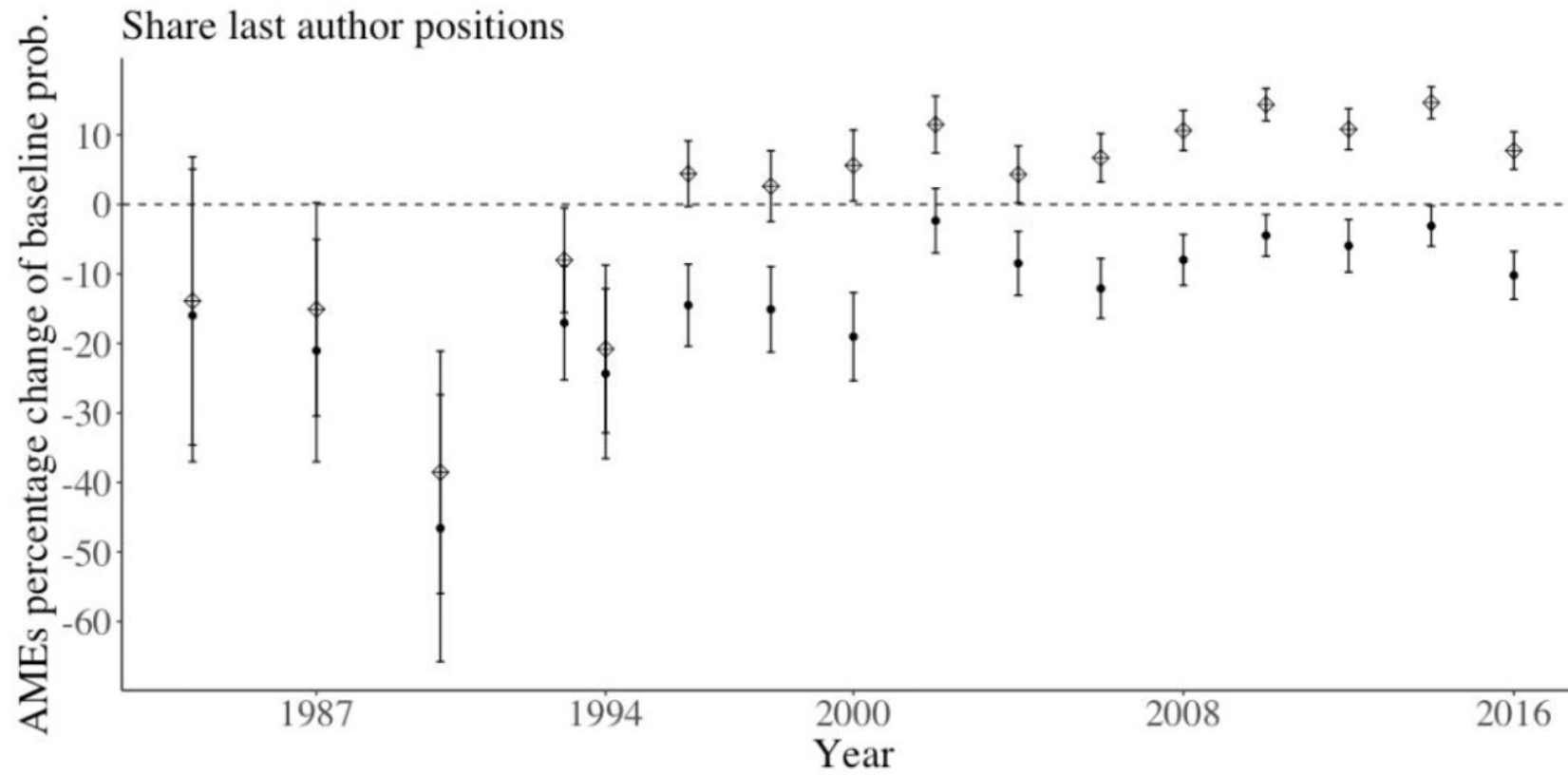
Results Inequality in Collaboration Networks over Time



Popularity According to Status over Time I

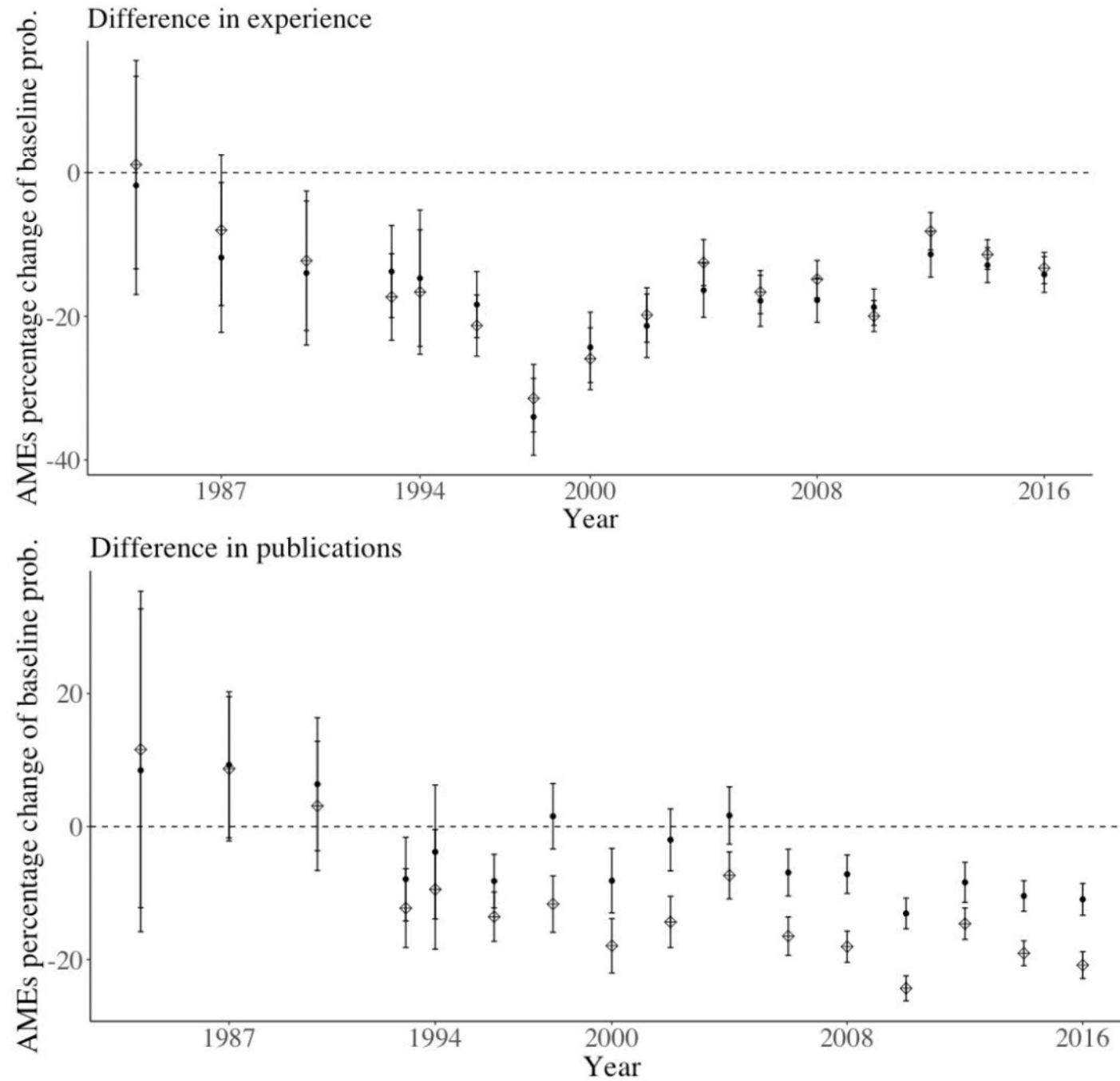
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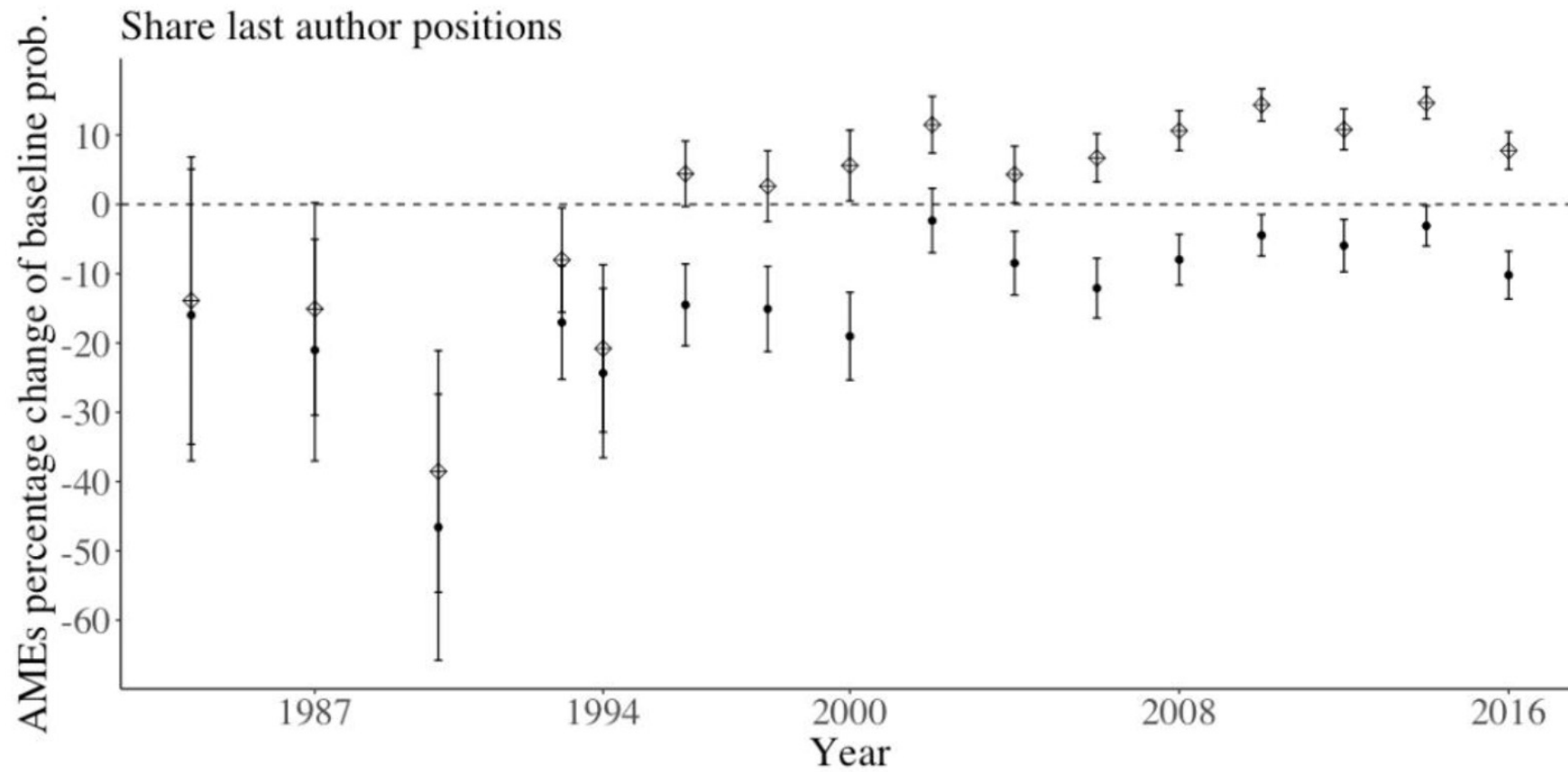
Segregation According to Status over Time II

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Segregation According to Status over Time II

19



Conclusion

- ▶ Inequality in social recognition increased with field size and maturity above and beyond the mechanical link between size and graph level indices
- ▶ Multiple status markers matter for researcher's popularity as the field evolves
- ▶ Segregation according to status differences increases over time
- ▶ Methodological contribution: application of AME to large networks of scientific collaboration for the first time, comparison of effect sizes across network models

Limitations

- ▶ Lack of information on classical socio-demographic characteristics
- ▶ No dissection of selection and influence regarding the role of status characteristics for network structure
- ▶ Individual outcomes not considered, how does the economy of attaining positions in the field change over time?
- ▶ Measurement does not allow for a test of different mechanisms that are related to increasing inequality (hoarding resources vs. cognitive explanation)
- ▶ Content of research not considered
- ▶ Investigation of one field, how does it generalize to other fields?

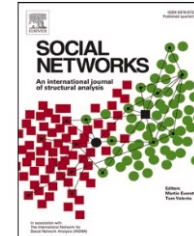
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Social Networks

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The development of stratification and segregation in a new scientific field: A study of collaboration among scientists in neuroblastoma research between 1975 and 2016[☆]

Mark Wittek^{a,*}, Christoph Bartenhagen^b, Frank Berthold^b

^a *Institute of Sociology and Social Psychology, University of Cologne, Meister-Ekkehardt-Straße 1, 50937 Köln, Germany*

^b *Department of Pediatric Oncology and Hematology, University of Cologne, Kerpener Str. 62, 50924 Köln, Germany*

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ABSTRACT

Using a new data set on scientific collaboration in neuroblastoma research over a period of 41 years, we study how the structure of collaboration ties in an evolving scientific field changes over time. Guided by concepts from the sociology of knowledge and status theories, we highlight the importance of a field's size and age for processes of stratification and segregation within it. Our findings suggest that stratification becomes stronger and diversifies as the field of neuroblastoma research expands. Moreover, we find that the occurrence of collaboration between researchers with a similar status becomes more likely as the field matures. While previous work has primarily examined established fields, our study focuses on how scientific fields change their structure over time. More broadly, our article demonstrates the potential analytical merits of adopting a dynamic and relational perspective on the developmental trajectories of organizational and cultural fields.

Thanks for your Attention!

Additional Slides

The Methods Odyssey

- ▶ **2019** Stochastic-Actor Oriented models (SAOMs; Snijders et al. 2011); abandoned because it's currently not legitimate to compare estimates between models of different networks (especially if the size changes); Tom Snijders is working on it though....
- ▶ **2020** Network regressions as simpler method; problem: similar issues in comparing coefficients (see logistic regressions Mood 2010)
- ▶ **2021** Simulations (cf. Bearman et al. 2004): simulating networks and study their global properties; cannot account for tie-formation mechanisms simultaneously, no solid statistical inference; advantage: you can get rid of the mechanical link between size and inequality (Anderson et al. 1999)
- ▶ **2022** Exponential random graph models (ERGMs; Lusher et al. 2013; Duxburry 2021); drawbacks: no separate parameters for selection influence, poor GOF, advantages: **allow for comparison of effect size over time**

Additional Descriptive Statistics

Table A1. Information on scientists' characteristics

Year	Years of experience					Number of publications per year					Seniority—share of last author positions				
	Mean	SD	Median	Min.	Max.	Mean	SD	Median	Min.	Max.	Mean	SD	Median	Min.	Max.
1975	0	0	0	0	0	0.69	1.65	0	0	6	0.15	0.38	0	0	1
1979	1.22	1.89	0	0	4	1.78	3.10	0.50	0	16	0.27	0.39	0	0	1
1984	1.73	2.89	0	0	9	0.45	1.20	0	0	7	0.17	0.35	0	0	1
1987	2.86	3.56	3	0	12	2.17	2.86	2.17	0	12	0.14	0.32	0	0	1
1990	2.02	3.63	0	0	15	1.96	2.67	1	0	11	0.19	0.36	0	0	1
1993	3.00	4.08	3	0	18	4.27	5.13	3	0	24	0.17	0.35	0	0	1
1994	2.23	3.34	1	0	19	3.76	5.63	2	0	28	0.16	0.33	0	0	1
1996	3.49	4.42	2	0	21	5.10	6.81	3	0	40	0.13	0.30	0	0	1
1998	3.66	4.51	3.66	0	23	6.18	8.58	3	0	63	0.14	0.30	0	0	1
2000	4.30	4.81	2	0	25	7.96	11.33	4	0	88	0.12	0.28	0	0	1
2002	4.30	5.10	4.30	0	27	7.56	12.62	3	0	110	0.12	0.27	0	0	1
2004	4.14	5.26	2	0	29	6.99	12.91	2	0	113	0.12	0.28	0	0	1
2006	4.69	5.53	2	0	31	7.37	13.83	3	0	125	0.13	0.30	0	0	1
2008	5.53	5.96	4	0	29	8.42	15.54	3	0	137	0.11	0.27	0	0	1
2010	5.51	6.33	4	0	31	8.35	16.46	3	0	155	0.10	0.25	0	0	1
2012	5.86	6.51	4	0	33	9.20	18.71	2	0	170	0.12	0.28	0	0	1
2014	6.64	6.86	4	0	35	9.67	19.95	3	0	183	0.09	0.25	0	0	1
2016	8.99	7.18	6	0	37	13.17	23.96	5	0	201	0.10	0.25	0	0	1

Figure A1. Difference in goodness of fit (GOF) between baseline and full model specification

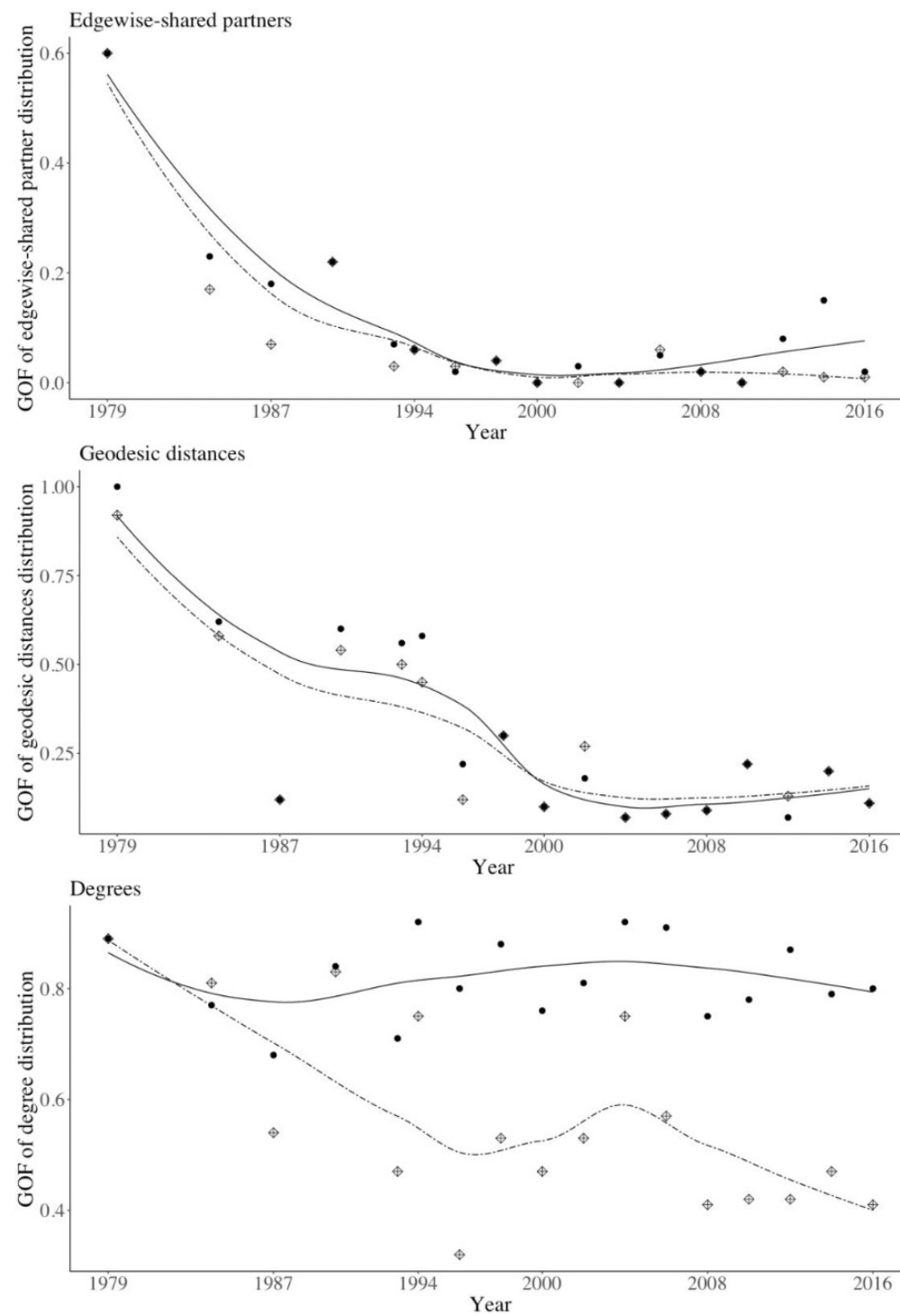


Table B1. Stochastic actor-oriented models SAOMs for collaboration network of neuroblastoma researchers

<i>Periods</i>	1975-1987	1984-1987	1987-1990	1990-1993	1993-1994	1994-1996	1996-1998	1998-2000
Density	-2.56*** (0.41)	-2.83*** (0.24)	14.18 (8.62)	-2.88*** (0.16)	10.15 (11.13)	-3.76*** (0.82)	-2.50*** (0.22)	-3.35*** (0.13)
Geometrically weighted edgewise-shared partners (GWESP)	5.90*** (1.00)	3.60*** (0.25)	10.56** (4.23)	4.10*** (0.20)	12.99 [†] (6.88)	5.09*** (0.46)	4.98*** (0.35)	4.67*** (0.14)
Degree activity (preferential attachment)	0.31*** (0.09)	0.19*** (0.04)	0.80 [†] (0.48)	0.17*** (0.02)	0.87 (0.50)	0.28* (0.15)	0.12*** (0.03)	0.11*** (0.01)
Degree homophily (assortativity)	-0.97*** (0.25)	-0.49*** (0.10)	-3.23 [†] (1.73)	-0.48*** (0.05)	-3.55 (2.20)	-0.75** (0.30)	-0.54*** (0.09)	-0.39*** (0.03)
Same country	-3.03*** (0.67)	-2.36*** (0.35)	-3.40 [†] (1.63)	-2.08*** (0.30)	-2.52 (2.19)	-1.73*** (0.35)	-1.09*** (0.13)	-2.11*** (0.15)
Same institution	1.08 [†] (0.64)	1.41*** (0.37)	-0.61 (1.38)	0.02 (0.26)	-0.62 (1.31)	-0.35 (0.44)	0.02 (0.14)	0.45*** (0.11)
Experience of author	0.02 (0.06)	0.07 [†] (0.037)	0.24* (0.12)	0.06 [†] (0.04)	-0.08 (0.10)	0.14* (0.07)	-0.03 (0.02)	0.01 (0.01)
Similar experience	0.08 (0.56)	0.43 (0.31)	1.66 (1.47)	0.83 (0.60)	-1.22 (1.49)	2.42** (0.80)	0.18 (0.30)	0.12 (0.27)
Publications of author (cumulative)	0.22** (0.08)	0.01 (0.13)	0.35** (0.13)	0.04 (0.04)	0.03 (0.06)	0.08** (0.03)	0.07*** (0.01)	0.03*** (0.01)
Similar number of publications	4.73*** (1.89)	0.42 (0.78)	2.34 (1.51)	0.44 (0.40)	-0.41 (1.56)	2.35*** (0.66)	1.67** (0.64)	2.16*** (0.40)
Share of last author positions	0.15 (0.42)	-0.21 (0.34)	-3.14 (2.32)	0.47* (0.24)	-3.82 (3.20)	-0.16 (0.48)	0.44* (0.21)	0.19 (0.20)
Similar share of last author positions	0.24 (0.39)	-0.26 (0.33)	0.17 (1.10)	0.22 (0.22)	-2.45 (2.65)	-0.27 (0.28)	0.26 (0.21)	0.17 (0.17)
Number of actors	158	152	240	323	364	531	708	721
Overall convergence ratio	0.16	0.17	0.14	0.17	0.15	0.17	0.17	0.14

<i>Periods</i>	2000-2002	2002-2004	2004-2006	2006-2008	2008-2010	2008-2012 ⁷⁶	2012-2014	2014-2016
Density	-2.09*** (0.29)	-1.37*** (0.21)	-3.59*** (0.11)	-4.93*** (0.14)	-5.83*** (0.42)	-5.97*** (0.18)	-5.70*** (0.74)	-7.96*** (0.51)
Geometrically weighted edgewise-shared partners (GWESP)	5.22*** (0.61)	5.86*** (0.18)	5.00*** (0.14)	4.43*** (0.09)	4.92*** (0.36)	4.99*** (0.12)	4.70*** (0.62)	5.17*** (0.33)
Degree activity (preferential attachment)	0.17*** (0.17)	0.20*** (0.02)	0.10*** (0.01)	0.04*** (0.004)	0.04*** (0.004)	0.05*** (0.003)	0.04*** (0.01)	0.03*** (0.003)
Degree homophily (assortativity)	-0.68*** (0.20)	-0.81*** (0.04)	-0.40*** (0.03)	-0.17*** (0.01)	-0.17*** (0.01)	-0.20*** (0.008)	-0.16*** (0.04)	-0.10*** (0.01)
Same country	-1.60*** (0.18)	-1.89*** (0.12)	-1.70*** (0.09)	-1.51*** (0.06)	-1.28*** (0.10)	-0.98*** (0.05)	-1.36*** (0.19)	-0.61*** (0.04)
Same institution	0.48*** (0.14)	0.26*** (0.09)	0.14* (0.075)	0.26*** (0.06)	0.45*** (0.07)	0.34*** (0.05)	0.42*** (0.05)	0.51*** (0.05)
Experience of author	0.05*** (0.01)	0.01 (0.01)	-0.02* (0.01)	-0.01* (0.005)	-0.007 [†] (0.004)	-0.007* (0.003)	-0.01*** (0.003)	0.01*** (0.003)
Similar experience of authors	1.02*** (0.32)	0.66* (0.28)	0.49* (0.23)	0.11 (0.14)	0.04 (0.10)	-0.02 (0.09)	-0.22 (0.22)	0.28** (0.11)
Publications of author (cumulative)	0.02* (0.01)	0.03*** (0.01)	0.04*** (0.005)	0.02*** (0.002)	0.02*** (0.002)	0.01*** (0.002)	0.02*** (0.002)	0.004*** (0.001)
Similar number of publications	0.87 (0.74)	0.76 (0.71)	3.69*** (0.49)	2.48*** (0.28)	3.02*** (0.24)	2.27*** (0.20)	3.24*** (0.63)	1.69*** (0.23)
Share of last author positions	-0.18 (0.23)	0.10 (0.22)	0.79*** (0.14)	0.44*** (0.11)	0.52*** (0.14)	0.46*** (0.09)	0.45** (0.18)	0.75*** (0.15)
Similar share of last author positions	-0.38 (0.24)	-0.07 (0.19)	0.39*** (0.13)	-0.05 (0.10)	0.15 (0.10)	0.15 [†] (0.08)	-0.002 (0.002)	0.08 (0.17)
Number of actors	814	1155	1257	1334	1517	1850	1585	1417
Overall convergence ratio	0.17	0.23	0.20	0.23	0.20	0.12	0.18	0.20

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